# Tony RuiKang OuYang

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## Education

**University of Cambridge** — *Cambridge, UK*
*PhD in Advanced Machine Learning* — *Oct 2025 – June 2028*
- Supervisor Prof. José Miguel Hernández-Lobato
- **Fully-funded** by EPSRC DLA Studentship Award

**University of Cambridge** — *Cambridge, UK*
*MPhil in Machine Learning and Machine Intelligence* — *Sep 2023 – Aug 2024*
- Graduate with **Distinction** (**Grade: 77%, Top 20%**)
- Course Advisor & Thesis Supervisor: Prof. José Miguel Hernández-Lobato
- Thesis: Energy-based Neural Sampler for Boltzmann Distribution (**Grade: 77.5%**)

**University of Oxford** — *Oxford, UK*
*Visiting Student in Mathematics and Statistics* — *Jan 2022 – Jul 2022*
- Supervisor: Prof. Gesine Reinert and Prof. Geoff Nicholls
- Fully funded by Harbin Institute of Technology, Shenzhen

**Harbin Institute of Technology, shenzhen** — *Shenzhen, CN*
*BEng in Data Science* — *Sep 2019 – Jun 2023*
- Graduate with **Distinction** (**Grade: 90%, Top 10%**)
- Thesis Supervisor: Yi Zhao
- Thesis: Implicit Local Node Embedding Synchronisation for Graph Autoencoders (**Grade: 90%**)

## Experiences

**AnewLabs** — **former Seed@ByteDance** — *July 2026 - present*
*Research Scientist Intern*
- Working on generative models for protein equilibrium sampling

**MolSS reading group** — *London, UK*
*Lead Organiser* — *May. 2025 - present*
- A community for machine learning in molecular science

**MLG@Cambridge** — *Cambridge, UK*
*Research Assistant in Probabilistic Machine Learning* — *Oct 2024 – Oct 2025*
Supervisor: Prof. José Miguel Hernández-Lobato

**The Alan Turing Institute** — *London, UK*
*Research Assistant* — *Apr. 2022 - Jul. 2022*
Supervisor: Prof. Gesine Reinert
- Research about scalable Graph Neural Network and Federated Learning for data privacy

## Publications

*Equal Contributions
- Grenioux, **OuYang**, and Wu (2026). Bridging Stochastic Flow Maps and Boltzmann Generators with Normalizing Flows. *Preprint.*
- **OuYang**, Yu, Ai, He, Boffi, Ravikumar, Hernández-Lobato, Simchowitz, Miller, and Chehab (2026). Few-Step Boltzmann Generators via Scalable Likelihood Flow Maps. *NeurIPS.* Also accepted at SPIGM@ICML.
- Yu, **OuYang**, Kaushik, Klami, Gutmann, and Chehab (2026). Learning Energy-Based Models from Stochastic Interpolants using Spatiotemporal Differences. *NeurIPS.*
- **OuYang***, Grenioux*, and Hernández-Lobato (2026). A Diffusive Classification Loss for Learning Energy-based Generative Models. *ICML.*
- Yu, **OuYang**, Horwood, and Hernández-Lobato (2026). Stochastic Interpolants in Hilbert Spaces. *Preprint.*
- **OuYang***, Rissanen*, He, Chen, Heinonen, Solin, and Hernández-Lobato (2025). Progressive Tempering Sampler with Diffusion. *ICML.*
- He*, Du*, Vargas, Zhang, Padhy, **OuYang**, Gomes, and Hernández-Lobato (2025). No Trick, No Treat: Pursuits and Challenges Towards Simulation-free Training of Neural Samplers. *FPI@ICLR.*

- **OuYang***, Qiang*, and Hernández-Lobato (2025). BNEM: A Boltzmann Sampler Based on Bootstrapped Noised Energy Matching. *TMLR*.
- **OuYang** (2024). Energy-Based Diffusion Neural Sampler for Boltzmann Densities. *MPhil Thesis*.
- **OuYang**, Elliott, Limnios, Cucuringu, and Reinert (2023). L2G2G: A Scalable Local-to-Global Network Embedding with Graph Autoencoders. *CNA*.

## Academic Services

- **journal reviewer**: TMLR (2026)
- **conference reviewer**: ICLR (2025–2027), ICML (2026), NeurIPS (2025–2026), AISTATS (2027)
  - **award:** silver reviewer @ ICML (2026), top reviewer @ NeurIPS (2026)
- **workshop reviewer**: DELTA@ICLR (2025–2026), FPI@NeurIPS (2025), SPIGM@NeurIPS (2025), DynaFront@NeurIPS (2025), SPIGM@ICML (2026), FoGen@ICML (2026), NonAR-LM@COLT (2026)
- **organizer**: MolSS reading group (Lead organizer)

## Supervision and Teaching

| **MEng Project Supervision** | | University of Cambridge |
|---|---|---|
| Sunaabh Trivedi | *Better Diffusion Models for Molecular Modelling* | *Oct 2024 - June 2025* |
| Luran Wang | *Towards better Neural Samplers for Boltzmann Distribution* | *Oct 2024 - June 2025* |
| **MPhil Thesis Supervision** | | University of Cambridge |
| Csaba Dékány | *Generative Models for Large-scale Materials* | *Apr 2026 - Aug 2026* |
| Ishaanvi Agrawal | *Generative Models for Large-Scale Materials* | *Apr 2026 - Aug 2026* |
| Shanzi Bao | *Exact-Density Flow Maps for Boltzmann Sampling* | *Apr 2026 - Aug 2026* |
| Antonio Franca | *Transferable Neural Transports for Parallel Tempering* | *Apr 2025 - Aug 2025* |
| James Yu | *Stochastic Interpolants in Hilbert Spaces* | *Apr 2025 - Aug 2025* |

## Awards and Achievements

**Distinction in MPhil Degree, University of Cambridge** *2024*
*Ranked top **20%** in MPhil in Machine Learning and Machine Intelligence*

**Travel and Research Grants, Wolfson College, University of Cambridge** *2023*
*Award for supporting my travelling for conference*

**Outstanding Graduates, Harbin Institute of Technology (Shenzhen)** *2023*
*Ranked top **10**% in HITsz*

**Distinction in BEng Degree, Harbin Institute of Technology (Shenzhen)** *2023*
*Ranked top **10**% in BEng in Data Science*

**International Exchange Program Scholarship, HITsz** *2022*
*Scholarship for visiting in Oxford, valued at abount **200,000** CNY*

## Others

**Athletic**

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|---|---|
| Player, Wolfson College (University of Cambridge) Football Varsity | *2023 - 2024* |
| Player, Harbin Institute of Technology (Shenzhen) Football Varsity | *2020 - 2023* |
| Captain, Harbin Institute of Technology (Shenzhen) Science School Football Team | *2021 - 2023* |
| 2nd place of HITsz Football League Season 22-23 | *2023* |
| 2nd place of HITsz Football League Season 20-21 | *2021* |
| 2nd place of HITsz Futsal League Season 20-21 | *2021* |
| 3rd place of HITsz Football League Season 19-20 | *2020* |

**Languages**

- Cantonese (1st Native)
- Chinese Mandarin (2nd Native)
- English (Professional working proficiency; IELT 7.0)
- French (Elementary proficiency)